

FEEDBACK AND CONTROL SYSTEMS

LABORATORY ACTIVITY #5

ZERO-POLE-GAIN MODELS AND INPUT RESPONSE

I. Learning Outcomes:

At the end of the laboratory activity, the students should be able to:

1. Use MATLAB in creating a zero-pole- gain models.
2. Plotting a zero-pole-gain models.
3. Using MATLAB for the impulse, step and arbitrary inputs in system response.

II. Laboratory Exercises:

1. Consider the transfer function

$$\frac{6s^2 + 1}{s^3 + 3s^2 + 3s + 7}$$

Using MATLAB plot the pole zero map of the above system.

2.

- a. Obtain the unit impulse response.

$$\frac{B(s)}{A(s)} = \frac{1}{s^2 + 0.2s + 1}$$

- b. Obtain the unit step response.

$$\frac{B(s)}{A(s)} = \frac{s}{s^2 + 0.2s + 1}$$

- c. Explain why the results in a and b are the same?

3. A system has a transfer function of

$$\frac{X(s)}{R(s)} = \frac{\left(\frac{15}{z}\right)(s + z)}{s^2 + 3s + 15}$$

Plot the response of the system when R(s) is a unit impulse and unit step for the parameter $z = 3, 6, 12$.

4. Consider the differential equation $y'' + 4y' + 4y = u$ where $y(0) = y'(0) = 0$ and $u(t)$ is a unit step. Determine the solution analytically and verify by co-plotting the analytical solution and the step response obtained with a step function.